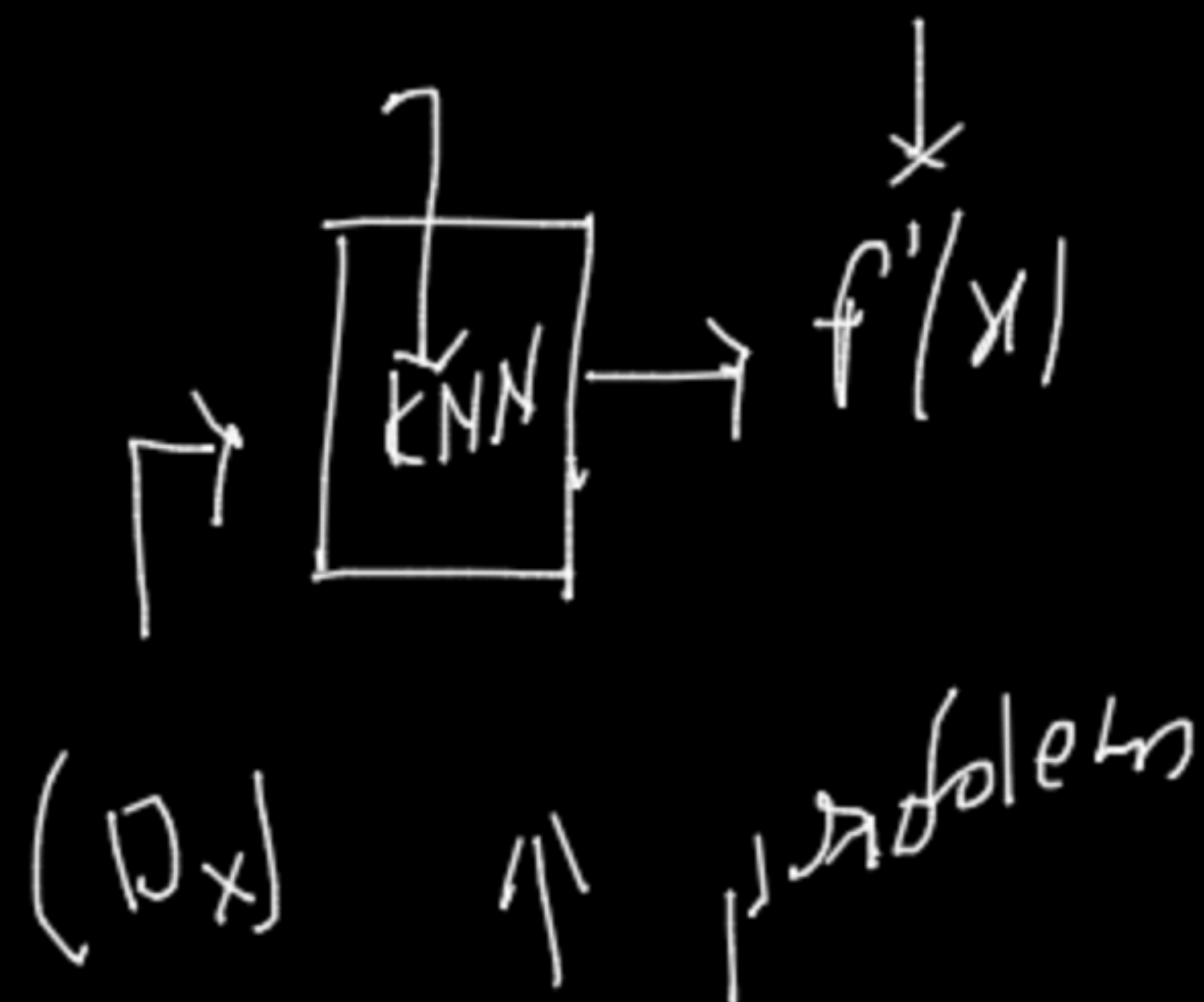


D_x

f_1	f_2	f_3	Y
-	-	-	-
-	-	-	-
-	-	-	-

$\rightarrow$ f_1 f_2 f_3
categorical col $\rightarrow$ Numerical col

Number $\rightarrow$ prediction



eg,

weight	hair-color	$Y \leftarrow$ height
45		
55		
65		
1		
100		

 $\uparrow$ Black $\rightarrow$ categorical data x

? →
categorical col → Numerical
? transformation

Black → 0 ← if you are saying
↑
WM

eg. translation → how are you?
transformations

justification →

all right?

know

↓

Rule

↓

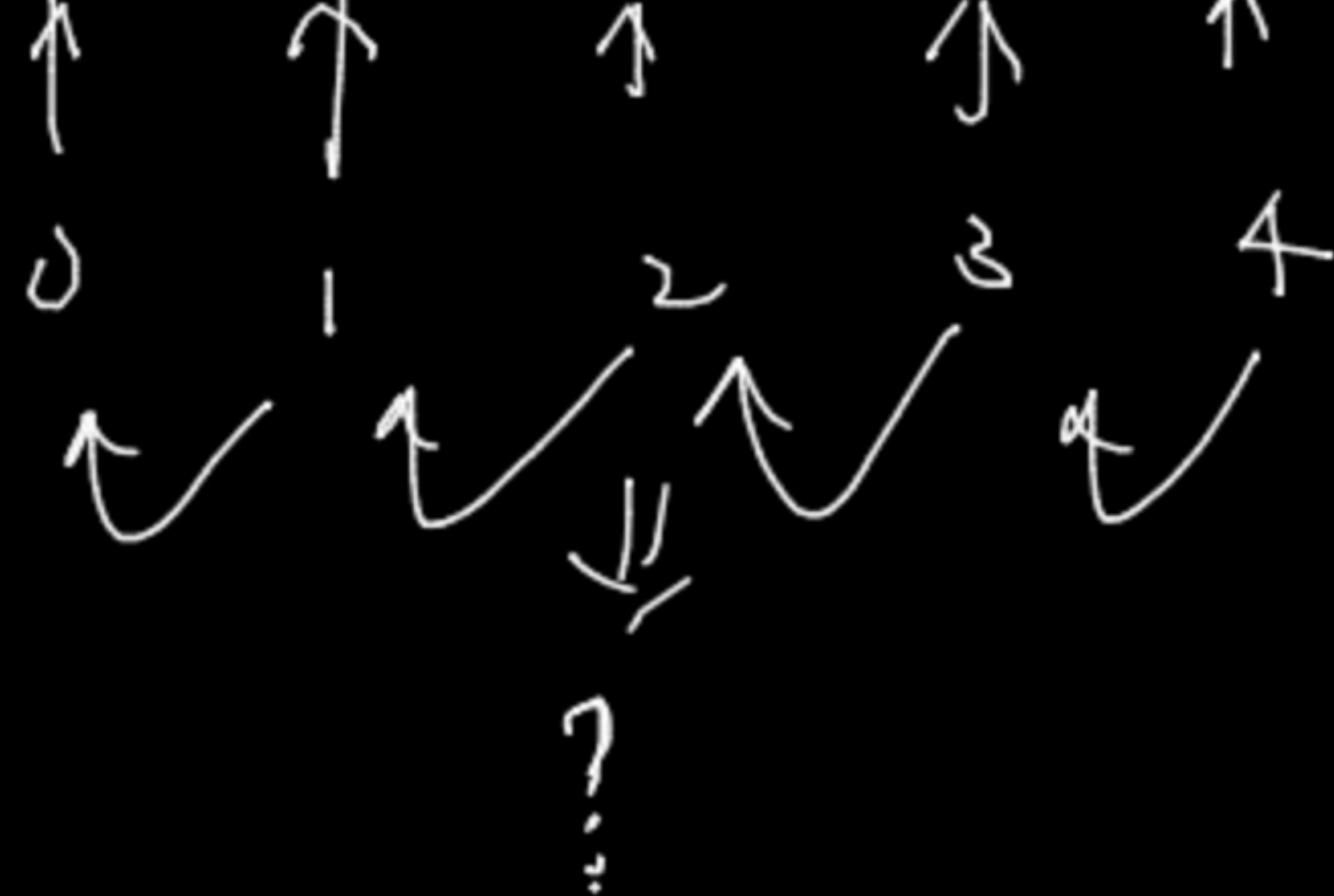
Program

meaning

↓

hair-color $\rightarrow \{BL, Br, Red, G, Grey\}$

Given $\rightarrow$



categorical

vector

(X)
KNN(
(X, Y)

"ONE-HOT ENCODING"

Data

BL	Br	Red	G	Grey
1	0	0	0	0

	weight	hair-color	
Review 1 good	45	Br 0	180
0 bad	47	BL 1	155
bad	48	Br 1	165
good	49	Br 1	175
	50	good 4	162
	65	whisk 3	177
2 very bad	45	Red 2	180

↑ numerical

↑ X =

train-color $\Rightarrow$ { Bl, Br, Red, G, Green } -

$\Rightarrow$

$\Rightarrow$ weight

train-color

$\uparrow$

weight

44

45

64

X

100

Bl

Br

Red

G

Green

Y

1

0

0

0

0

180

1

0

0

0

0

165

0

1

0

0

0

185

KNN(x, y)

numerical

$\uparrow$

trained

20

$\rightarrow$

60

$\overline{\text{one-hot}}$

$\rightarrow$

one-hot

$\uparrow$

$$As^{4+}$$
$$\begin{array}{r} 250 \\ \times 4 \\ \hline 1000 \end{array}$$

△ ↑

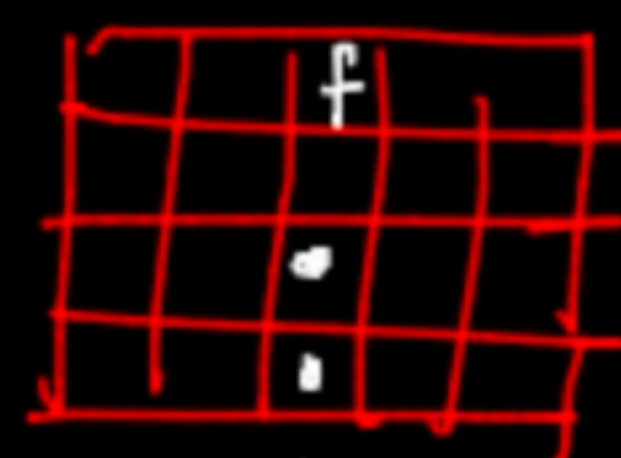
Missing value imputation

DE team

eg. Retail

Raw Data

pre-process



mean
median
mode
PS
Basic

ML/DL

yes
no
why
why
'0'
+
no
why
sense?

X			Y
Age	income		
(23)	(14)		yes
(24)	15		yes
(26)	16		yes
(25)	15.5		no
(26)	15.5		no
Null			

fail
KNN (X, Y)
(26.5)
(20-30)
(23, 14, 26, 15.5)
4
14+15+16
3
15.5

Banking

lid	income
C1	14
C2	14
C3	15
C4	15

credit score

~~NA~~ (-1)

650 ↑ domain

655 ↑

670

↑ (data)

→ mean = 650 + 655 + 670 / 3

Domain

DS

(data) (notes)

{col 1
col 2
col 3
col 4}

'EDA'

↓

imp

④

1()

↑

LMN

← pre-processing

Blackboard